CST575 - Theory of Computation

CST 575 (Theory of Computation) Syllabus

Course Information

Course Details

- **Course Title:** Theory of Computation
- **Course Number:** CST 575
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course introduces the fundamental theoretical foundations of computer science, focusing on formal languages, automata theory, and computability. Students will explore regular languages, context-free languages, Turing machines, and the concepts of decidability and undecidability.

Materials

- Course textbook: "Introduction to the Theory of Computation" by Michael Sipser

Technology Requirements

Students must have access to: - A computer with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Analyze formal languages and their classifications
2. Construct and evaluate finite automata, pushdown automata, and Turing machines
3. Understand the limits of computation and undecidability
4. Apply theoretical models to practical computational problems

Grading

Assignment Kind	Value
Homework & Problem Sets	40%
Exams	40%
Participation	20%

Homework & Problem Sets

Homework assignments will involve problem sets on automata, formal proofs, and computational theory. Students will be expected to demonstrate problem-solving and proof-writing skills.

Exams

Exams will test theoretical knowledge, including formal proofs, construction of automata, and analytical reasoning.

Participation

Participation includes active involvement in classroom problem-solving, discussions, and collaboration with peers.

Grade Assignment

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion. Students will be notified of any substantive changes via Canvas announcement and/or email.